

Statistics (II): Regression analysis DATA130046.01

Time: 13:00-15:00, 26 December, 2022;

Answer ALL FIVE questions. Marks are shown in square brackets.

1. A study was conducted to determine the joint effects of temperature and concentration of herbicide on absorption of 2 herbicides on a commercial charcoal material. The data are given in the table. There were 2 observations for each treatment.

Temp	Concentration	
	20	40
10	0.28	0.38
10	0.278	0.392
55	0.266	0.332
55	0.258	0.334

(a) Consider a linear regression model of the absorption on temperature and concentration (as quantitative independent variables).

(i) Write down the regression model. State the assumptions. **[3 marks]**

(ii) Find the least square estimates. **[3 marks]**

(iii) Obtain the ANOVA table and hence test whether there is a regression between the absorption and the 2 independent variables at 5% level of significance. **[4 marks]**

(b) Consider a 2-way ANOVA model with interaction.

(i) Write down the 2-way ANOVA model. State the assumptions. Find the parameter estimates. **[3 marks]**

(ii) Obtain the ANOVA table. Is there a regression between the dependent variable and the 2 factors at the 5% level of significance? **[4 marks]**

(iii) Test whether or not the two factors interact; use $\alpha = 0.05$. State the alternatives, decision rule and conclusion. **[3 marks]**

2. For the one-way ANOVA model

$$y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, \dots, a, \quad j = 1, \dots, n_i,$$

where $\varepsilon_{ij} \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$. For given $c_1, \dots, c_a$, propose a test statistic for

$$H_0 : \frac{\mu_1}{c_1} = \dots = \frac{\mu_a}{c_a},$$

then find its null distribution and rejection region. **[20 marks]**

3. Consider the linear regression model

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

We make a prediction at a new point x_0 based on a ridge regression estimator with parameter λ : $\hat{Y} = x_0^T \hat{\beta}^{\text{ridge}}(\lambda)$. Derive explicit expressions for the bias, variance and MSE of $\hat{Y}$ as a function of λ . **[20 marks]**

4. Consider the signal-plus-noise model such that

$$\mathbf{Y} = \mu + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

- (1). Find the BLUE of μ and its MSE. **[6 marks]**
 (2). Consider another shrinkage estimator $\hat{\mu} = (\hat{\mu}_1, \dots, \hat{\mu}_n)'$ such that

$$\hat{\mu}_i = Y_i (1 - a / \|\mathbf{Y}\|^2).$$

Find the MSE of $\hat{\mu}$. **[6 marks]**

- (3) Find a such that the MSE of $\hat{\mu}$ is smaller than the MSE of the BLUE. [Hint: using the following Stein's lemma: If $X \sim N(\mu, \sigma^2)$ then $E[(X - \mu)h(X)] = \sigma^2 E[h'(X)]$ with $h(X)$ continuous.] **[8 marks]**

5. Ridge maximum likelihood estimates of the logistic model parameters are found by the maximization of the ridge penalized loglikelihood

$$\begin{aligned} L(Y, X; \beta, \lambda) &= L(Y, X; \beta) - \frac{1}{2} \lambda \|\beta\|^2 \\ &= \sum_{i=1}^n \left\{ Y_i X_i^T \beta - \log(1 + \exp(X_i^T \beta)) \right\} - \frac{1}{2} \lambda \beta^T \beta, \end{aligned}$$

where the second summand is the ridge penalty (the sum of the square of the elements of β) with λ the penalty parameter. Find the algorithm (the one similar to the iteratively re-weighted least squares algorithm for logistic regression) and give the updating step changes. **[20 marks]**

$\alpha=0.05$

F α k1 k2	1	2	3	4	5	6	8	12	24	∞
1	161.4	199.5	215.7	224.6	230.2	234.0	238.9	243.9	249.0	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.41	19.45	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.84	8.74	8.64	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.04	5.91	5.77	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.68	4.53	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.15	4.00	3.84	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.73	3.57	3.41	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.28	3.12	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.23	3.07	2.90	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.91	2.74	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	2.95	2.79	2.61	2.40
12	4.75	3.88	3.49	3.26	3.11	3.00	2.85	2.69	2.50	2.30
13	4.67	3.80	3.41	3.18	3.02	2.92	2.77	2.60	2.42	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.70	2.53	2.35	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.64	2.48	2.29	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.59	2.42	2.24	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.55	2.38	2.19	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.51	2.34	2.15	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.48	2.31	2.11	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.28	2.08	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.42	2.25	2.05	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.40	2.23	2.03	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.38	2.20	2.00	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.36	2.18	1.98	1.73
25	4.24	3.38	2.99	2.76	2.60	2.49	2.34	2.16	1.96	1.71
26	4.22	3.37	2.98	2.74	2.59	2.47	2.32	2.15	1.95	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.30	2.13	1.93	1.67
28	4.20	3.34	2.95	2.71	2.56	2.44	2.29	2.12	1.91	1.65
29	4.18	3.33	2.93	2.70	2.54	2.43	2.28	2.10	1.90	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.09	1.89	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.00	1.79	1.51
60	4.00	3.15	2.76	2.52	2.37	2.25	2.10	1.92	1.70	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.02	1.83	1.61	1.25
∞	3.84	2.99	2.60	2.37	2.21	2.09	1.94	1.75	1.52	1.00